

Sabhyata Sinha

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Education

Manipal University Jaipur

B.Tech in Computer Science and Engineering (Artificial Intelligence & Machine Learning)

Aug 2023 – Jul 2027

CGPA: 7.96

Experience

Tiara Rides (AIC-MUJ Hybrid) — *AI/ML Intern*

May 2025 – Nov 2025

- Reduced fare prediction error by **40% (RMSE)** through feature engineering on vehicle and usage attributes.
- Deployed real-time inference services using **FastAPI REST APIs** and interactive dashboards with Streamlit.
- Integrated **C++ inference modules** with Python pipelines for low-latency predictions.
- Developed a face recognition system using **OpenCV** for driver verification and fraud prevention.

Amazon ML Summer School 2026 — *Machine Learning Summer School*

Jul 2026 – Aug 2026

- Successfully completed all modules of Amazon's competitive **Machine Learning Summer School 2026**, led by Amazon Scientists and Engineers.
- Completed advanced training in **supervised learning, deep neural networks, dimensionality reduction, unsupervised learning, sequential learning, reinforcement learning, Generative AI, LLMs, and causal inference**.
- Strengthened practical expertise in **machine learning model development, evaluation, and production-oriented AI workflows**.

Projects

AG-USegNet: Hybrid Attention Network for Brachial Plexus Ultrasound Segmentation — Deep Learning, TensorFlow, ONNX, C++ Deployment

[GitHub](#)

- Designed and iteratively enhanced an attention-guided ultrasound nerve segmentation framework, evolving from the lightweight **LAG-Net** architecture to the proposed **AG-USegNet** hybrid **U-Net + SegNet** model with attention-gated skip connections.
- Achieved an **F1-Score of 0.8856** and **IoU of 0.7947**, delivering a **25% improvement** over baseline segmentation models.
- Developed an end-to-end **Python → ONNX → C++** deployment pipeline for near real-time inference and benchmarked the model against **Attention U-Net, DeepLabV3+, TransUNet, and SwinSeg**.

PayRecover AI: AI-Powered Payment Recovery — Machine Learning, Python, Scikit-learn, Pandas, Streamlit

[GitHub](#)

- Built an AI-powered payment recovery system that predicts recovery probability, estimates expected recovery value, and prioritizes failed payments for targeted recovery actions.
- Engineered **34 features** from **50,000 payment records** and evaluated multiple machine learning models, including **Logistic Regression, Random Forest, Extra Trees, and HistGradientBoosting**.
- Achieved **98.51% recall, 88.02% F1-score, 0.7165 ROC-AUC, and 0.8903 PR-AUC** on the held-out test set using a tuned decision threshold.
- Developed a rule-based **decision engine** and interactive **Streamlit dashboard** to recommend automatic retry, personalized recovery, manual review, or stopping automated recovery based on predicted recovery probability and payment value.

Skills

Languages: Python, C++, JavaScript, SQL

ML/AI: Machine Learning, Deep Learning, Computer Vision, CNNs, Transformers, LLMs, Feature Engineering, Model Evaluation, MLOps

Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV

Libraries/Tools: NumPy, Pandas, ONNX, FastAPI, Streamlit, Git, Linux, Jupyter, REST APIs

Core CS: Data Structures & Algorithms, Object-Oriented Programming, Image Segmentation, Model Deployment

Coding Profiles

LeetCode: [200+ Problems](#) — [NeetCode 150](#)

Achievements

- Ranked **4918** in **Goldman Sachs India Hackathon 2026**.
- Qualified for the **Women Who Master 2026** Zonal Round.
- Advanced to **Round 2** of **TechMCode 2026**.
- Advanced to **Round 2** of **EY Techathon 6.0**.
- Selected for the **McKinsey Forward Program 2026**.
- Open Source Contributor at **SWOC** and **PWOC**.